

Week 1 – Introduction of Financial Risk Management

Welcome to Financial Risk Management. In week 1, we have two tasks; one for reviewing basic statistics and mathematics and modern portfolio theory, the other for introducing financial risk management. There are two PowerPoint files for reviewing both basic statistics and mathematics and modern portfolio theory. Please review two review files and solve the practice problems for quiz #1. Then finish quiz #1.

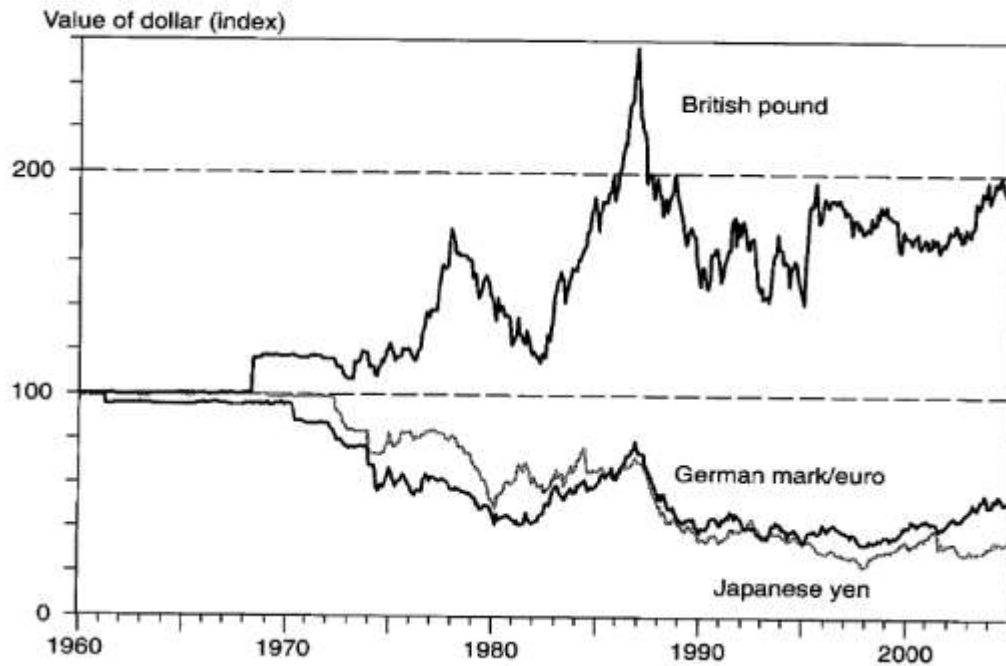
In this week 1 lecture note, I will motivate you to study financial risk management. It is likely to be a summary of chapters 1 through 3 in *Value at Risk* by Jorion.

1.1 Financial Risk

What is risk? As you learnt from Corporate Finance, which is a pre-requisite of this course, risk is unexpected outcomes, and measured by standard deviation. Firm's risk is basically consisting of business risk and financial risk.

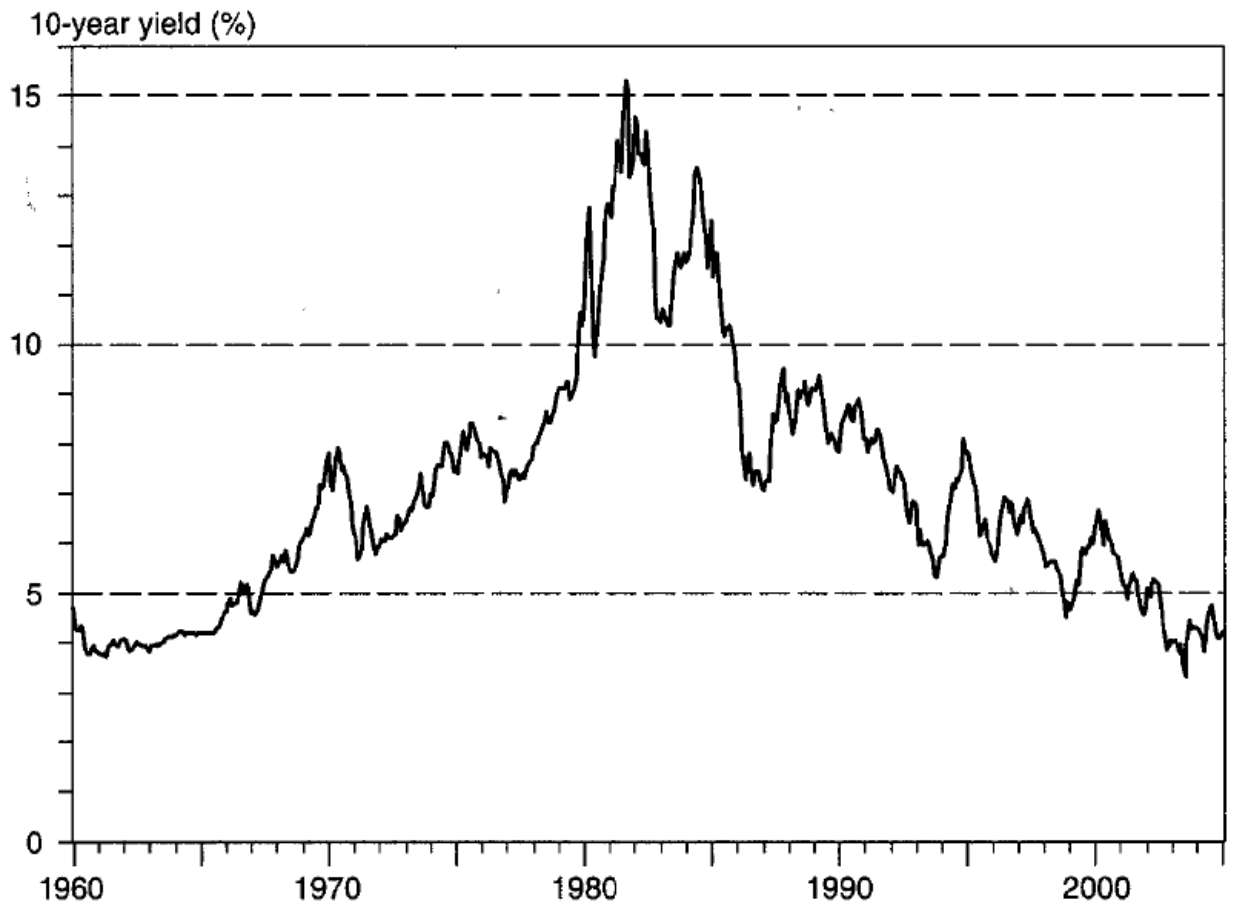
- Firm's risk
 - Business Risk: the company assumes to create a competitive advantage and add to value for shareholders. It's the risk the company willing to face to run the business.
 - Financial Risk: Possible losses owing to financial market activities, resulting from interest rate movement, commodity price changes, and changes in the stock market prices.
- New trend: Increased volatility of financial markets

Movements in the dollar.



As you see in the figure above, the fixed exchange system broke down in 1971, resulting in flexible and volatile exchange rate.

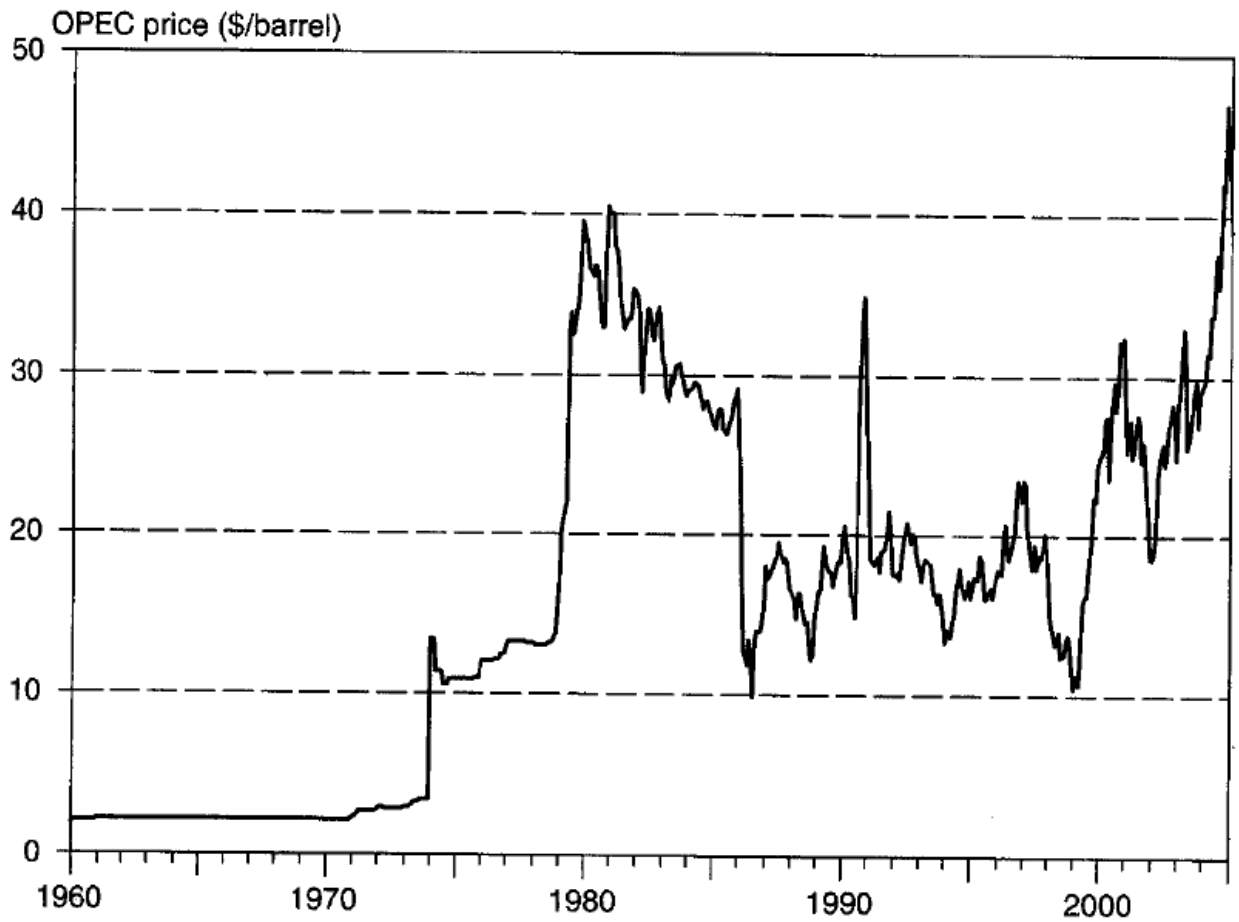
Movements in U.S. interest rates.



The figure above shows 10-year Treasury bond yield, which is one of the most important benchmarks for many financial instruments, for example 30-year fixed rate mortgage. It clearly indicates the volatile interest rate.

How about commodity prices? Let's look at the oil prices.

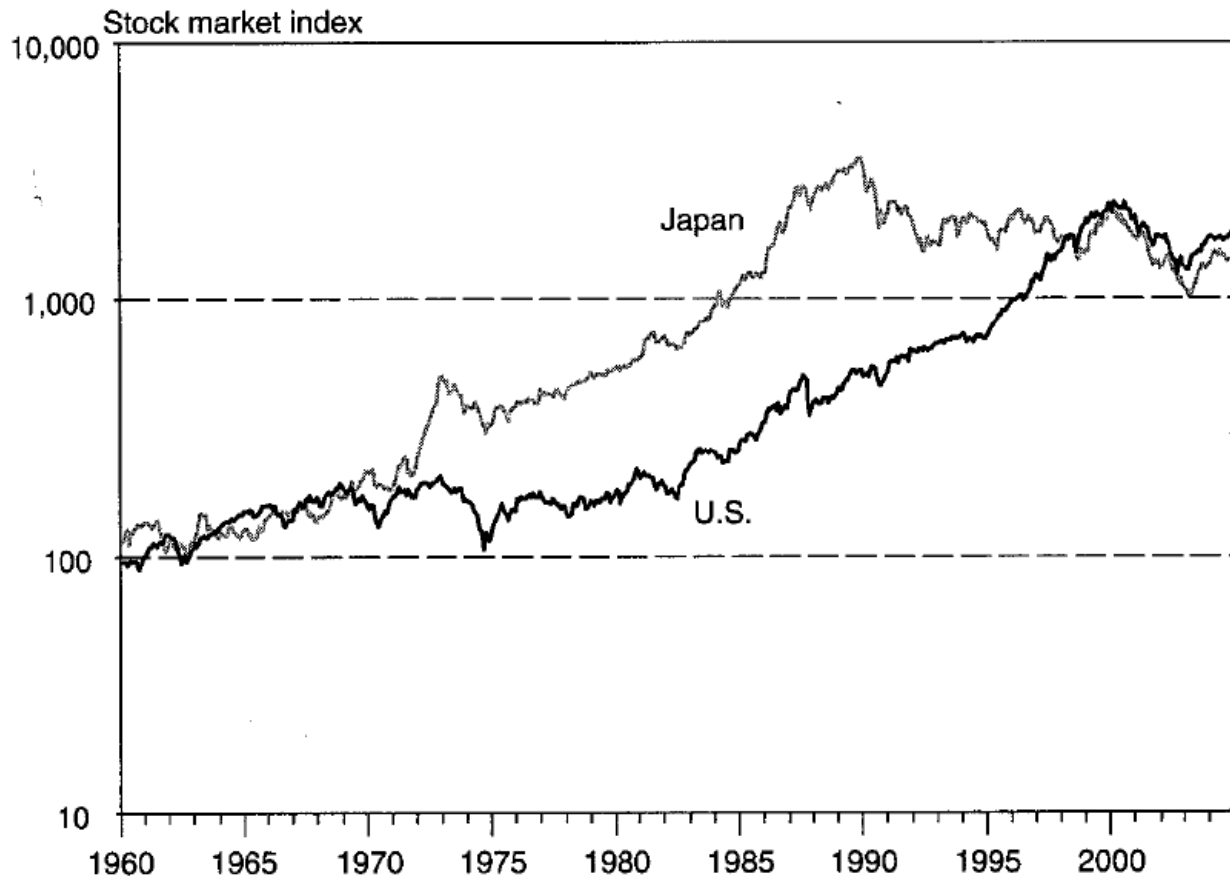
Movements in oil prices.



After two oil shocks in the 1970s, this important commodity price changes dramatically, even nowadays.

How about stock markets?

Movements in stock prices.



The major stock market prices both in the US and Japan show significant volatility throughout 1990s and 2000s.

Financial risk management refers to the design and implementation of procedures for identifying, measuring, and managing financial risks. After financial crises in 1990s the Value at Risk (VaR) was developed to measure financial risk. Now what is VaR?

1.2 What is VaR?

Value at risk (VaR) is a method that attempts to provide a single number summarizing the total market risk in a portfolio of financial assets (contrary to a huge number of partial risk measures such as “Greek letters analysis”)

VaR is intended to express the expected maximum loss (worst loss) over a target horizon within a given confidence level.

- Building blocks of VaR

- Probability distribution: normal distribution
- Target horizon

$$t\text{-day VaR} = 1\text{-day VaR} \times \sqrt{t}$$

- Confidence level

Confidence level is a percentage number that determines how far the portfolio should be protected against unexpected losses (i.e. Basel II uses 99% confidence level). A confidence level C determines a cut-off point α_C (quantile) such that the area to its right captures $C\%$ of all probability and consequently the area to its left captures a residual $(1-C)\%$ of all probability:

$$P(\tilde{R} > \alpha_C) = C\% \text{ or equivalently } P(\tilde{R} \leq \alpha_C) = (1 - C)\%$$

- Example

A portfolio has a value of 100 mil USD and its average monthly return is 3.5%. There is only a 5% chance that the portfolio will fall by more than 1.7% over a month. Therefore the value at risk for 95% confidence level is:

$$VaR(95\%) = [0.035 - (-0.017)] \times 100\text{mil} = 5.2 \text{ mil USD}.$$

The amount 5.2 mil USD represents a monthly underperformance relative to the average outcome that happens with 5% probability.

- VaR can be used:

- Information reporting (senior managers are informed about the risks run by trading and investment operations)
- Resource allocation (setting position limits for trades)
- Performance evaluation (benchmark for proper evaluation of risk exposure)
- Prudential regulation (maintaining minimum levels of capital reserves against financial risks)

Now let's move on what we mean by financial risk.

1.3 Types of financial risks

- Market risk: the risk of losses owing to movements in the level or volatility of market prices. Market risk includes financial variables such as stock prices, interest rates, exchange rates, and commodity prices.
- Liquidity risk: It arises when a transaction cannot be conducted at prevailing market prices owing to thin markets.
- Credit risk: The risk from the counterparties who cannot fulfill their contractual obligations.
- Operational risk: The risk resulting from inadequate or failed internal processes, people and systems or from external events.

1.4 Lessons from financial disasters

Do we have any episodes which reflect the financial risks? We, actually have many. The detailed description is in the Jorion's textbook.

- Barings' Rogue Trader
- Metallgesellschaft
- Orange County
- Daiwa's Lost Billion
- AIB's Cost Savings

After these well-publicized losses the followings are regulators' response.

1.5 VaR-Based regulatory capital

- Risk-based capital is a trend
- Regulation of commercial banks
 - Basel Capital Accord of 1988 (BASEL I)
 - Only credit risks
 - Minimum amount of capital
 - Tier 1 capital and tier 2 capital
 - At least 8% of the total risk-weighted assets
 - The Basel II Accord (2004)
 - Three pillars
 - 1) Minimum regulatory requirements (risk based capital requirements): Credit, market and operational risks 8% of risk-weighted capital

- 2) Supervisory review
 - 3) Market discipline
- Capital > TRC = CRC + MRC + OPC
- Bank assets
 - 1. Trading book – proprietary trading
 - 2. Banking book – loans
- Standardized method
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- Internal model
 - Quantitative parameters
 - 10 trading days
 - 99% CI
 - 1 year of historical data
 - Correlation
 - Market risk charge
 - Plus factor: a penalty component

$$MRC_t^{IMA} = \max(k \frac{1}{60} \sum_{i=1}^{60} VaR_{t-i}, VaR_{t-1}) + SRC_t$$

Now we know that the VaR method is well received both from financial industry and financial regulators. Starting next week we will learn how to calculate VaR from real data.